15-441 Computer Networking

Lecture 5 - Coding and Error Control

Peter Steenkiste

Eric Anderson

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www.cs.cmu.edu/~prs/15-441-F13

From Signals to Packets

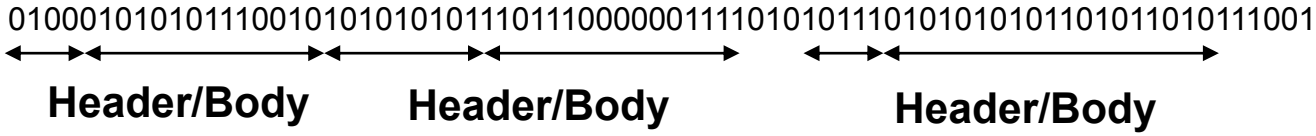
Analog Signal

"Digital" Signal

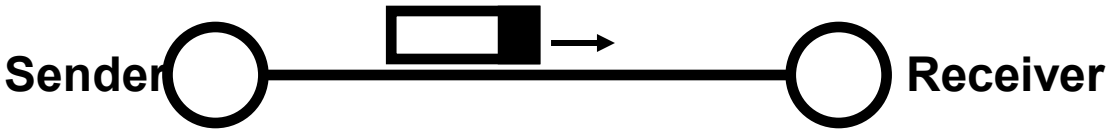
Bit Stream

0 0 1 0 1 1 1 0 0 0 1

Packets

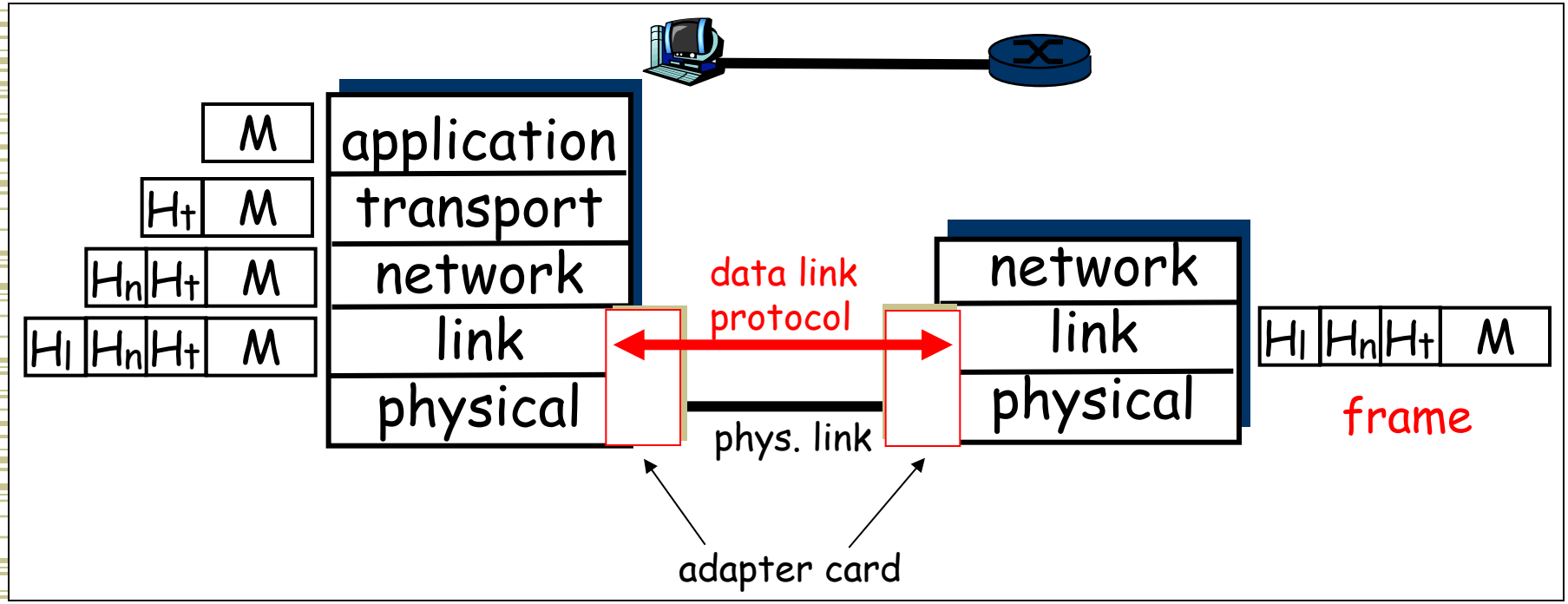


Packet Transmission



Link Layer: Implementation

- Implemented in "adapter"
  - E.g., PCMCIA card, Ethernet card
  - Typically includes: RAM, DSP chips, host bus interface, and link interface



Datalink Functions

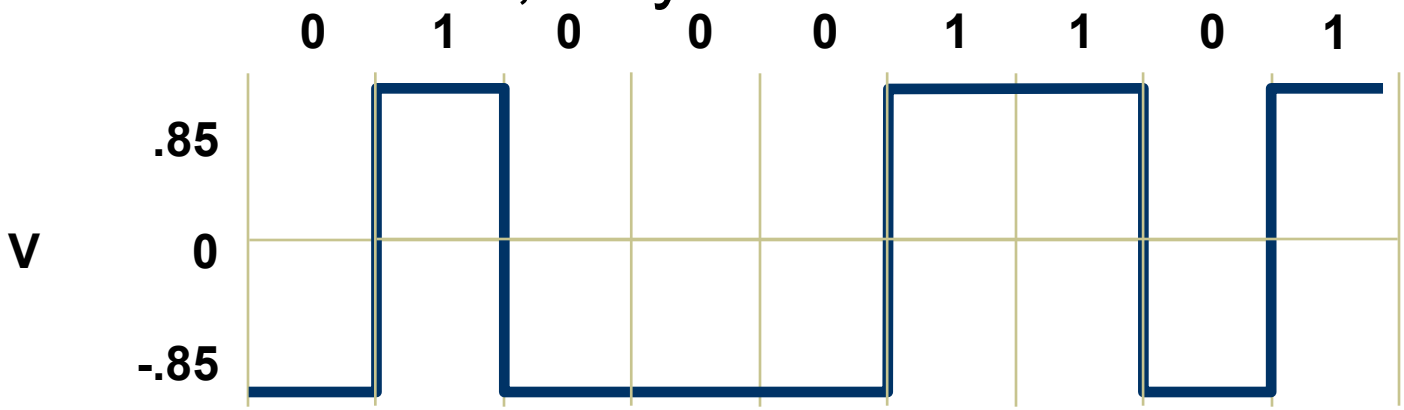
- Framing: encapsulating a network layer datagram into a bit stream.
  - Add header, mark and detect frame boundaries
- Media access: controlling which frame should be sent over the link next.
- Error control: error detection and correction to deal with bit errors.
  - May also include other reliability support, e.g. retransmission
- Flow control: avoid that the sender outruns the receiver
- Hubbing, bridging: extend the size of the network

Outline

- Encoding and decoding
  - Translate between bits and digital signal
- Framing
  - Bit stream to packets
- Packet loss & corruption
  - Error detection
  - Flow control
  - Loss recovery

How Encode?

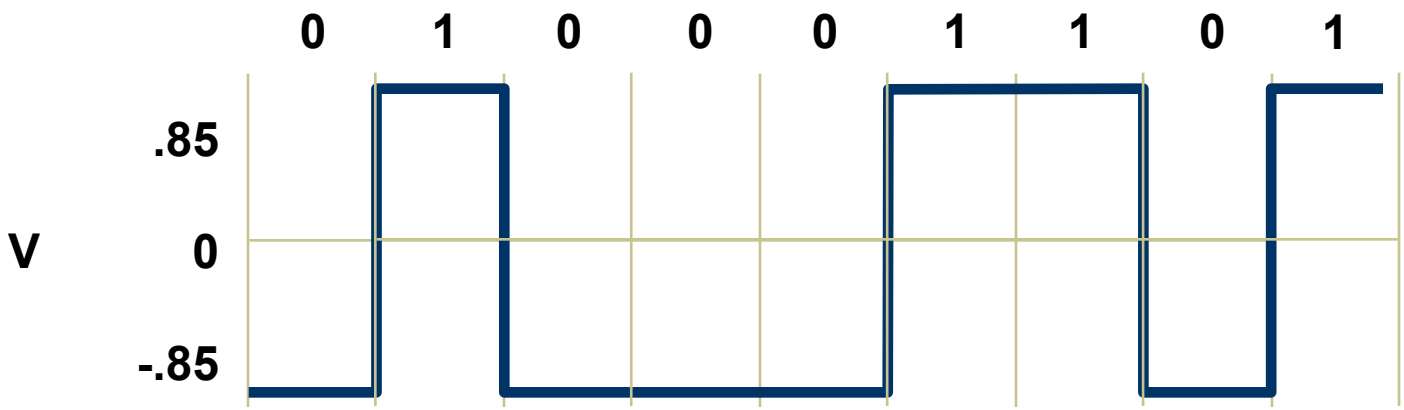
- Seems obvious, why take time with this?



Why Do We Need Encoding?

- Keep receiver synchronized with sender.
- Create control symbols, in addition to regular data symbols.
  - E.g. start or end of frame, escape, ...
- Error detection or error corrections.
  - Some codes are illegal so receiver can detect certain classes of errors
  - Minor errors can be corrected by having multiple adjacent signals mapped to the same data symbol
- Encoding can be done one bit at a time or in multi-bit blocks, e.g., 4 or 8 bits.
- Encoding can be very complex, e.g. wireless.

Non-Return to Zero (NRZ)

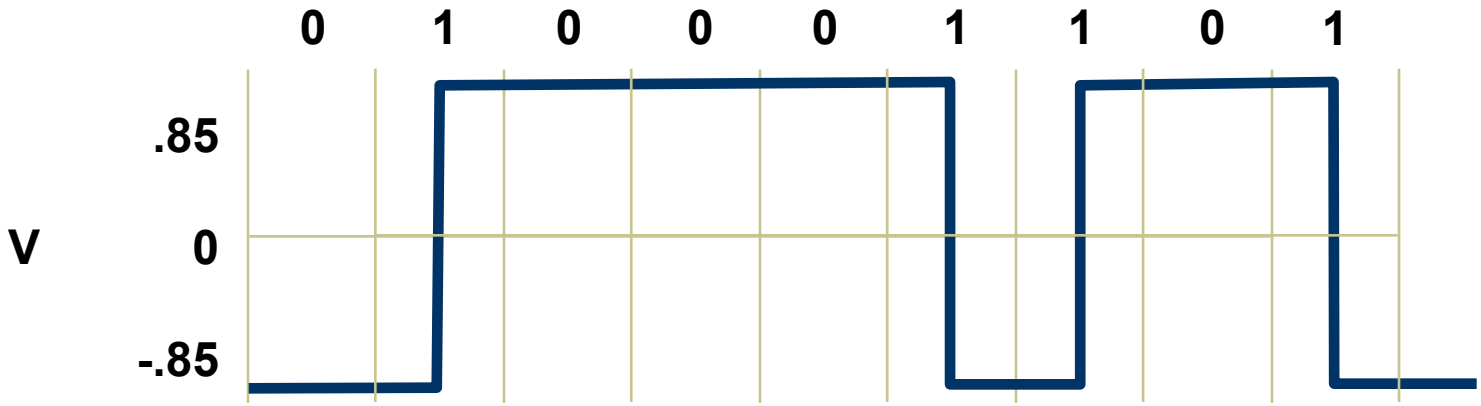


- 1 → high signal; 0 → low signal
- Used by Synchronous Optical Network (SONET)
- Long sequences of 1's or 0's can cause problems:
  - Sensitive to clock skew, i.e. hard to recover clock
  - DC bias hard to detect – low and high detected by difference from average voltage

Clock Recovery

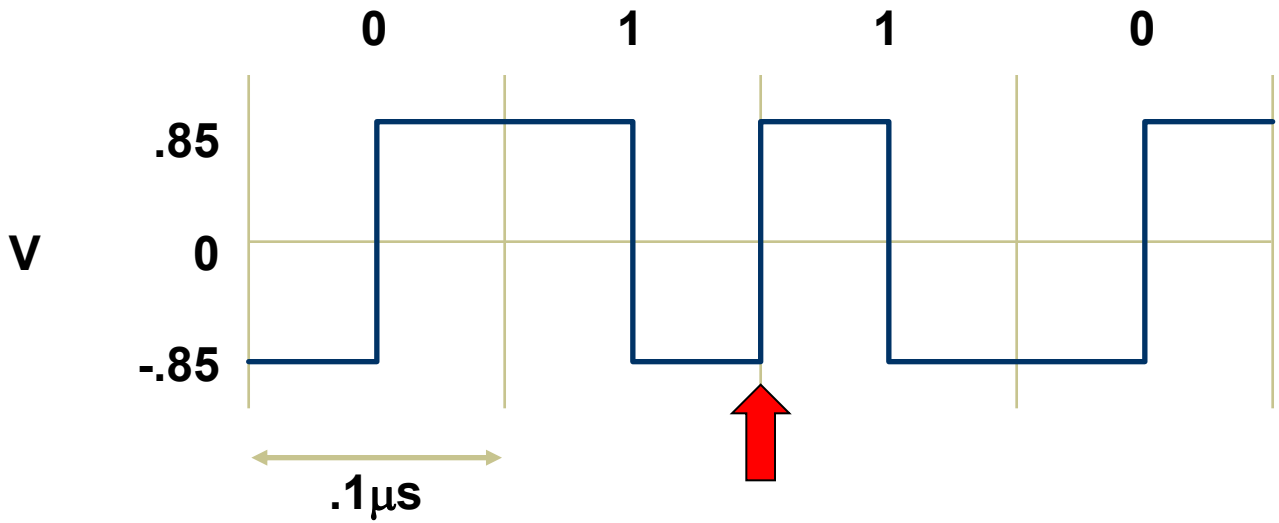
- When to sample voltage?
- Synchronized sender and receiver clocks
- Need easily detectible event at both ends
  - Signal transitions help resync sender and receiver
  - Need frequent transitions to prevent clock skew
  - SONET XOR's bit sequence to ensure frequent transitions

Non-Return to Zero Inverted (NRZI)



- 1 → make transition; 0 → signal stays the same
- Solves the problem for long sequences of 1's, but not for 0's.

Manchester Encoding



- Used by Ethernet
- 0=low to high transition, 1=high to low transition.
- Transition for every bit simplifies clock recovery
- DC balance has good electrical properties
- But you pay a price …
  - Doubles the number of transitions – more spectrum!
  - Circuitry must run twice as fast

4B/5B Encoding

- Data coded as symbols of 5 line bits → 4 data bits, so 100 Mbps uses 125 MHz.
  - Uses less frequency space than Manchester encoding
- Encoding ensures no more than 3 consecutive 0's
- Uses NRZI to encode resulting sequence
- 16 data symbols, 8 control symbols
  - Data symbols: 4 data bits
  - Control symbols: idle, begin frame, etc.
- Example: FDDI.

4B/5B Encoding

Data	Code
0000	11110
0001	01001
0010	10100
0011	10101
0100	01010
0101	01011
0110	01110
0111	01111

Data	Code
1000	10010
1001	10011
1010	10110
1011	10111
1100	11010
1101	11011
1110	11100
1111	11101

Other Encodings

- 8B/10B: Fiber Channel and Gigabit Ethernet
- 64B/66B: 10 Gbit Ethernet (& 40 and 100 Gb/S)
- B8ZS: T1 signaling (bit stuffing)

Things to Remember

- Encoding necessary for clocking
- Lots of approaches
- Rule of thumb:
  - Little bandwidth → complex encoding
  - Lots of bandwidth → simple encoding

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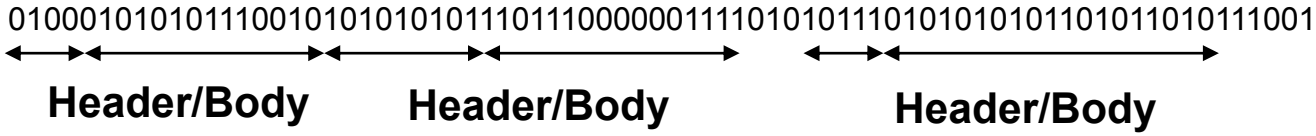
"Digital" Signal

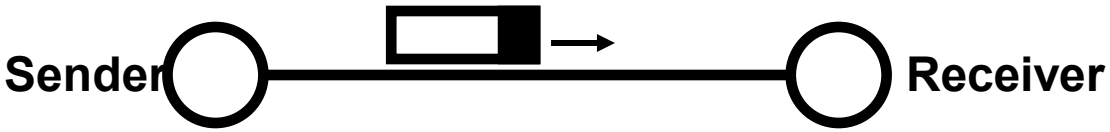
Bit Stream

0 0 1 0 1 1 1 0 0 0 1

Packets



Packet Transmission



Outline

- Encoding
  - Digital signal to bits
- Framing
  - Bit stream to packets
- Packet loss & corruption
  - Error detection
  - Flow control
  - Loss recovery

Framing

- How do we break up a stream of bits into frames?

0100010101011100101010101011101110000001111010101110101010101101011010111001

Framing

- A link layer function, defining which bits have which function.
- Minimal functionality: mark the beginning and end of packets (or frames).
- Some techniques:
  - out of band delimiters (e.g. 4B/5B control symbols)
  - frame delimiter characters with character stuffing
  - frame delimiter codes with bit stuffing
  - synchronous transmission (e.g. SONET)

Out-of-band: E.g., 802.5

- 802.5/token ring uses 4b/5b
- Start delim & end delim are “illegal” data codes

Delimiter Based

- SYN: sync character
- SOH: start of header
- STX: start of text
- ETX: end of text

- What happens when ETX is in Body?

Character and Bit Stuffing

- Mark frames with special character.
  - What happens when the user sends this character?
  - Use escape character when controls appear in data:
  - *abc*def → *abc*def
  - Very common on serial lines, in editors, etc.
- Mark frames with special bit sequence
  - must ensure data containing this sequence can be transmitted
  - example: suppose 11111111 is a special sequence.
  - transmitter inserts a 0 when this appears in the data:
  - 11111111 → 111111101
  - must stuff a zero any time seven 1s appear:
  - 11111110 → 111111100
  - receiver unstuffs.

Ethernet Framing

- Preamble is 7 bytes of 10101010 (5 MHz square wave) followed by one byte of 10101011
- Allows receivers to recognize start of transmission after idle channel

Clock-Based Framing

- Used by SONET
- Fixed size frames (810 bytes)
- Look for start of frame marker that appears every 810 bytes
- Will eventually sync up

How avoid clock skew?

- Special bit sequences sent in first two chars of frame
  - But no bit stuffing. Hmmm?
- Lots of transitions by xoring with special pattern (and hope for the best)

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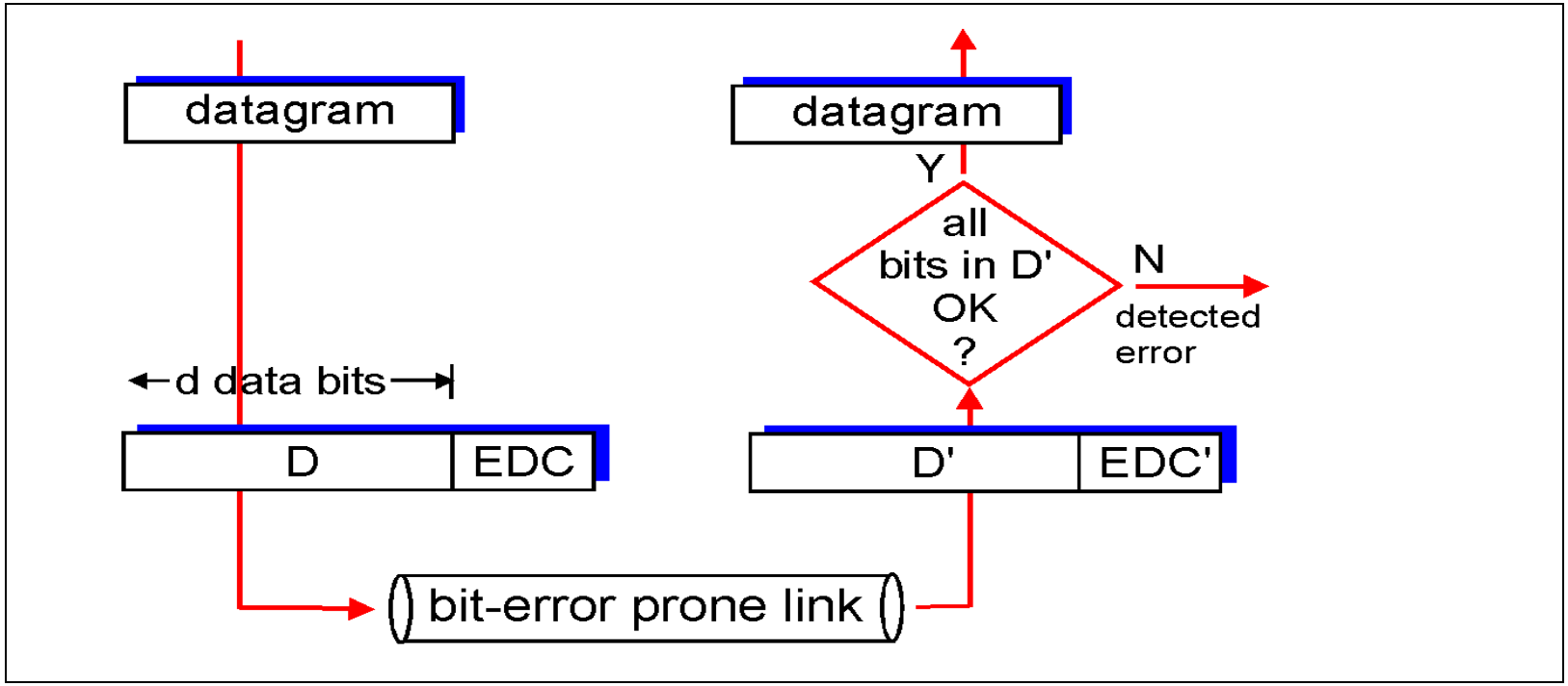
Error Coding

- Transmission process may introduce errors into a message.
  - Single bit errors versus burst errors
- Detection:
  - Requires a convention that some messages are invalid
  - Hence requires extra bits
  - An (n,k) code has codewords of n bits with k data bits and r = (n-k) redundant check bits
- Correction
  - Forward error correction: many related code words map to the same data word
  - Detect errors and retry transmission

Error Detection

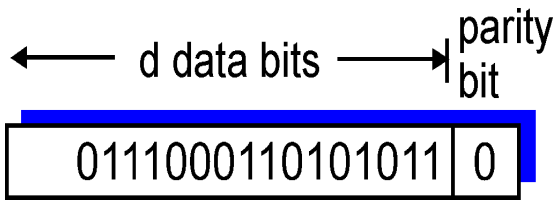
- EDC= Error Detection and Correction bits (redundancy)
- D = Data protected by error checking, may include header fields
- Error detection not 100% reliable!
  - Protocol may miss some errors, but rarely
  - Larger EDC field yields better detection and correction



Parity Checking

Single Bit Parity:

Detect single bit errors



Internet Checksum

- Goal: detect “errors” (e.g., flipped bits) in transmitted segment

Sender

- Treat segment contents as sequence of 16-bit integers
- Checksum: addition (1’s complement sum) of segment contents
- Sender puts checksum value into checksum field in header

Receiver

- Compute checksum of received segment
- Check if computed checksum equals checksum field value:
  - NO - error detected
  - YES - no error detected. But maybe errors nonethless?

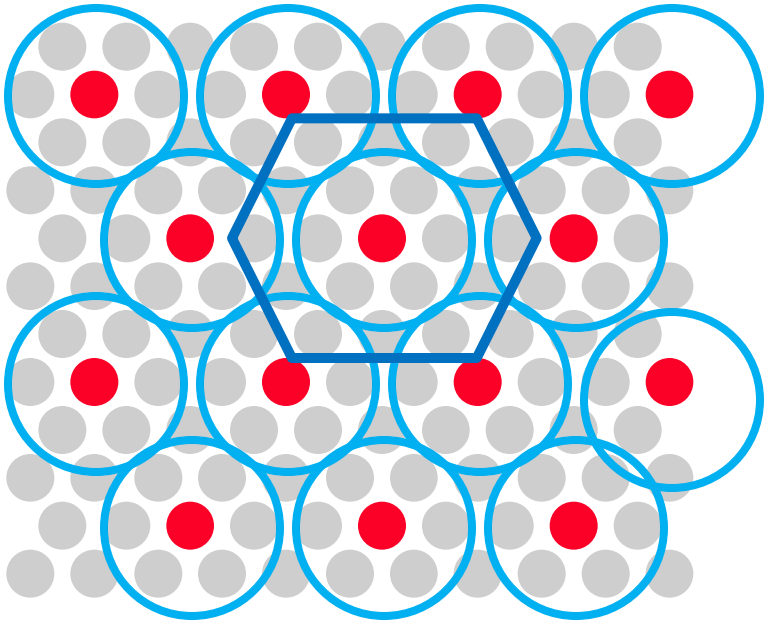
Basic Concept: Hamming Distance

- Hamming distance of two bit strings = number of bit positions in which they differ.
- If the valid words of a code have minimum Hamming distance D, then D-1 bit errors can be detected.
- If the valid words of a code have minimum Hamming distance D, then [(D-1)/2] bit errors can be corrected.

1	0	1	1	0
1	1	0	1	0

HD=2

HD=3



Cyclic Redundancy Codes (CRC)

- Commonly used codes that have good error detection properties.
  - Can catch many error combinations with a small number of redundant bits
- Based on division of polynomials.
  - Errors can be viewed as adding terms to the polynomial
  - Should be unlikely that the division will still work
- Can be implemented very efficiently in hardware.
- Examples:
  - CRC-32: Ethernet
  - CRC-8, CRC-10, CRC-32: ATM

CRC: Basic idea

- Treat bit strings as polynomials:

  1 0 1 1 1

  $X^{4+} \quad X^2+X^1+X^0$

- Sender and Receiver agree on a *divisor* polynomial of degree k
- Message of M bits → send M+k bits
- No errors if M+k is divisible by divisor polynomial
- If you pick the right divisor you can:
  - Detect all 1 & 2-bit errors
  - Any odd number of errors
  - All Burst errors of less than k bits
  - Some burst errors >= k bits

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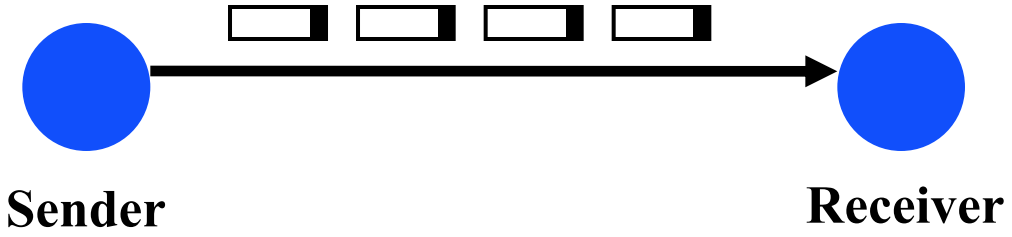
Link Flow Control and Error Recovery

- Dealing with receiver overflow: flow control.
- Dealing with packet loss and corruption: error control.
- Meta-comment: these issues are relevant at many layers.
  - Link layer: sender and receiver attached to the same “wire”
  - End-to-end: transmission control protocol (TCP) - sender and receiver are the end points of a connection
- How can we implement flow control?
  - “You may send” (windows, stop-and-wait, etc.)
  - “Please shut up” (source quench, 802.3x pause frames, etc.)
  - Where are each of these appropriate?

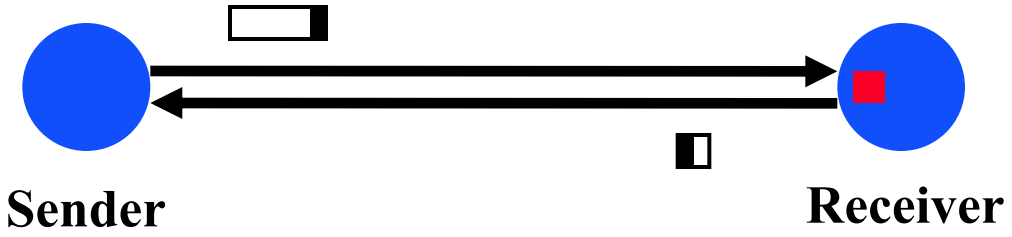
A Naïve Protocol

- Sender simply sends to the receiver whenever it has packets.
- Potential problem: sender can outrun the receiver.
  - Receiver too slow, buffer overflow, ..
- Not always a problem: receiver might be fast enough.

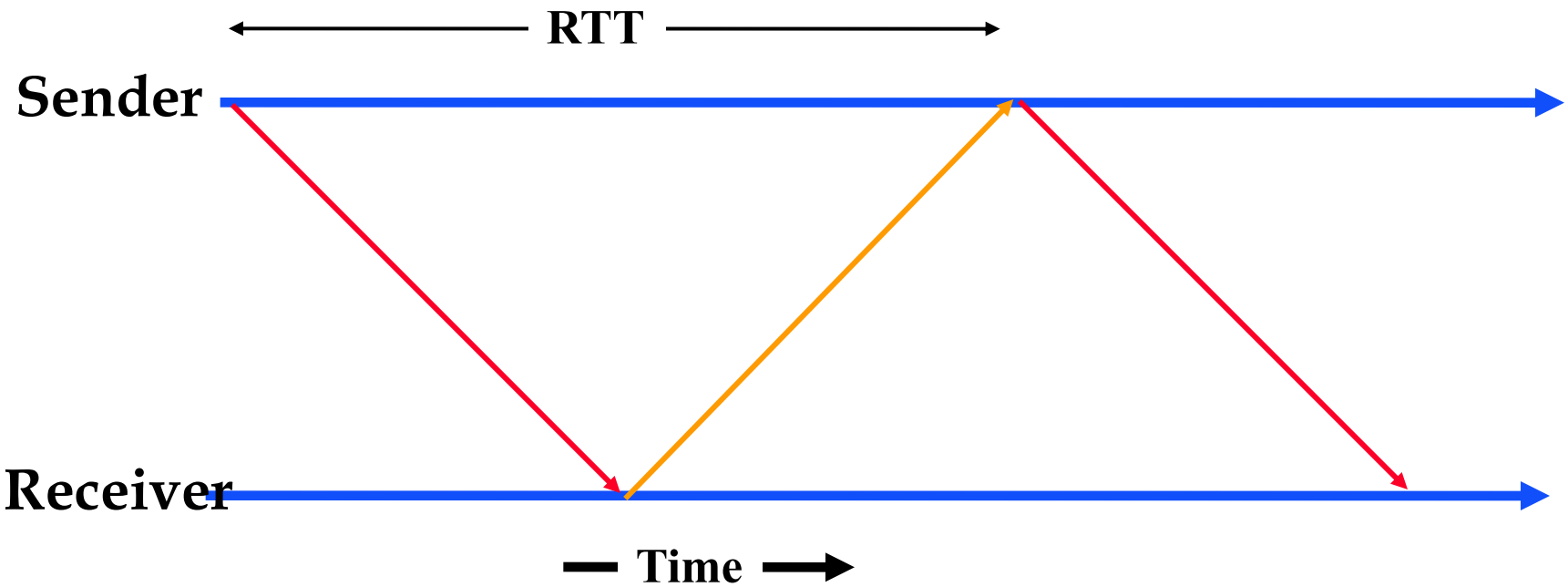


Adding Flow Control

- Stop and wait flow control: sender waits to send the next packet until the previous packet has been acknowledged by the receiver.
  - Receiver can pace the receiver



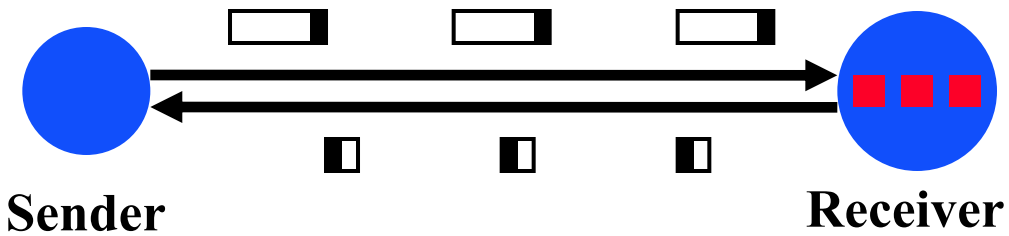
Drawback: Performance



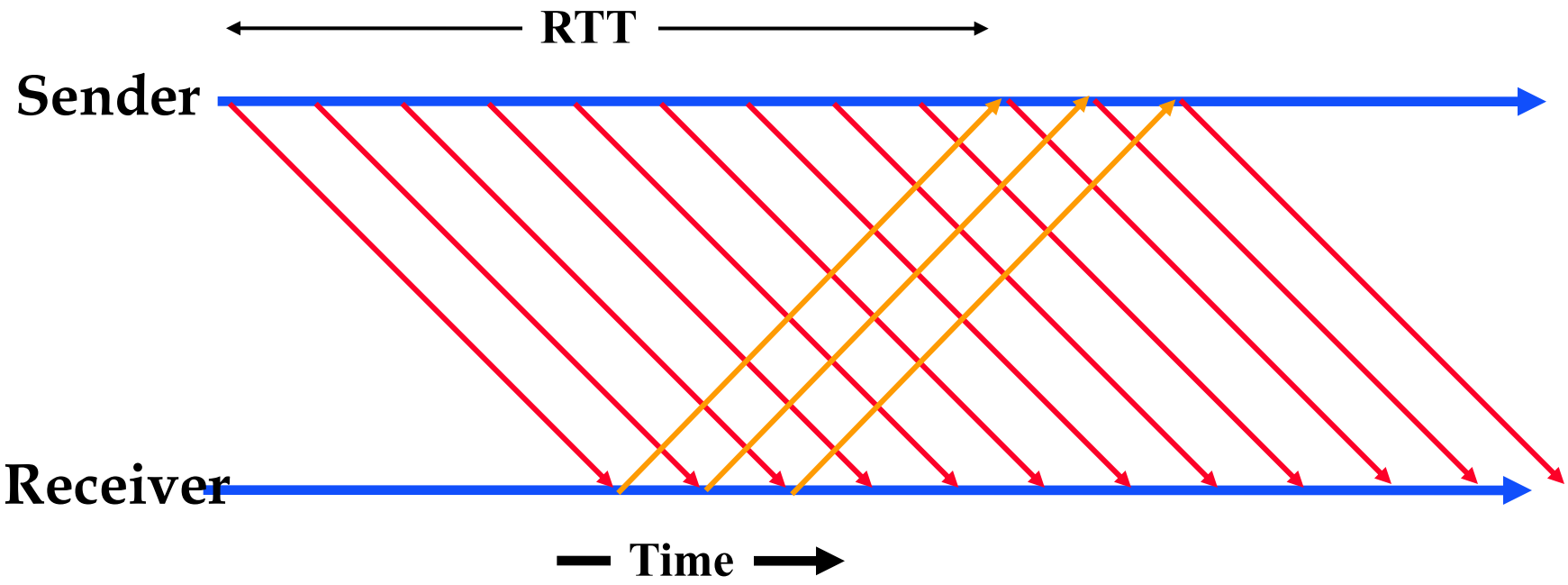
$$\text{Max Throughput} = \frac{1 \text{ pkt}}{\text{Roundtrip Time}}$$

Window Flow Control

- Stop and wait flow control results in poor throughput for long-delay paths: packet size/ roundtrip-time.
- Solution: receiver provides sender with a window that it can fill with packets.
  - The window is backed up by buffer space on receiver
  - Receiver acknowledges the a packet every time a packet is consumed and a buffer is freed



Bandwidth-Delay Product



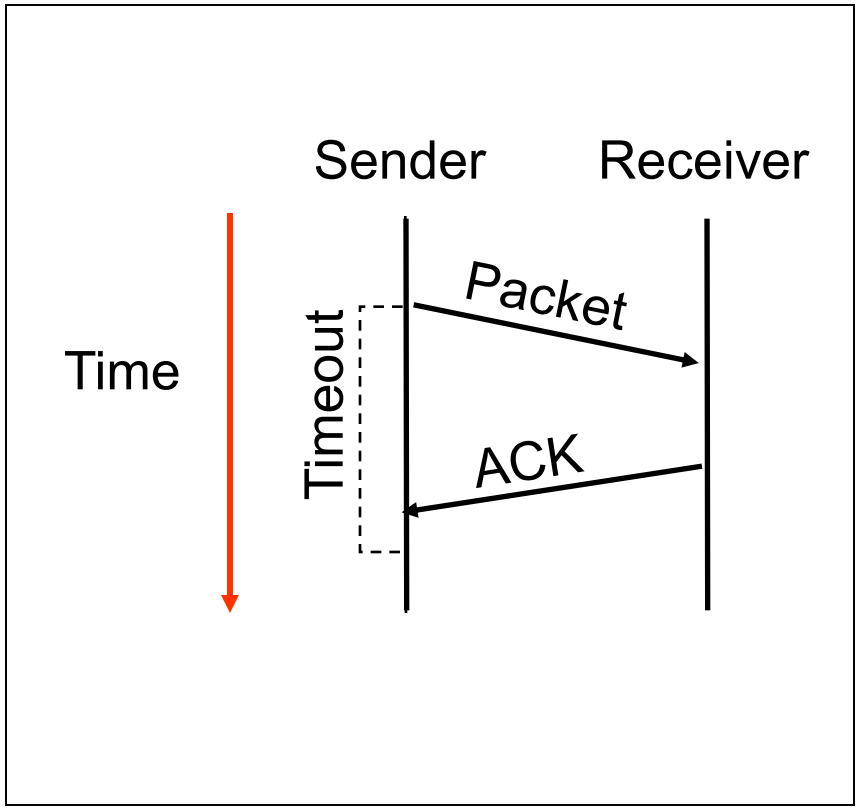
$$\text{Max Throughput} = \frac{\text{Window Size}}{\text{Roundtrip Time}}$$

Error Recovery

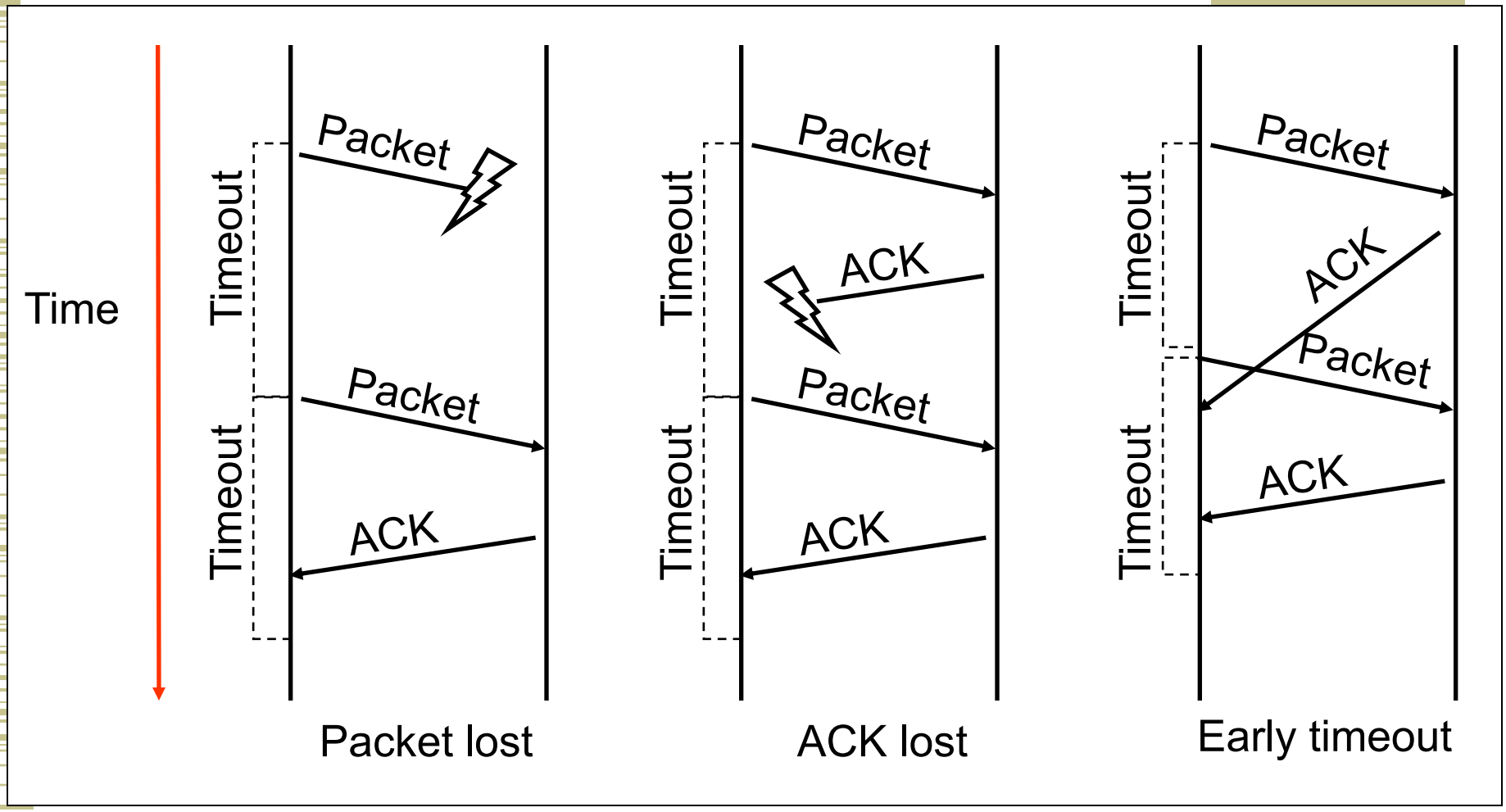
- Two forms of error recovery
  - Error Correcting Codes (ECC)
  - Automatic Repeat Request (ARQ)
- ECC
  - Send extra redundant data to help repair losses
- ARQ
  - Receiver sends acknowledgement (ACK) when it receives packet
  - Sender uses ACKs to identify and resend data that was lost
- Which should we use? Why? When?

Stop and Wait

- Simplest ARQ protocol
- Send a packet, stop and wait until acknowledgement arrives
- Will examine ARQ issues later in semester

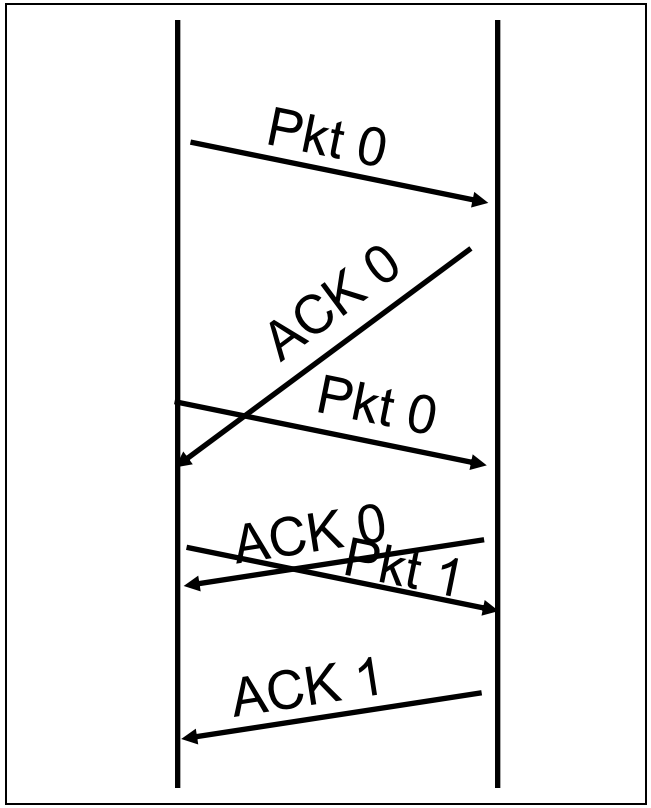


Recovering from Error



How to Recognize Retransmissions?

- Use sequence numbers
  - both packets and acks
- Sequence # in packet is finite → How big should it be?
  - For stop and wait?
- One bit – won't send seq #1 until received ACK for seq #0



Implementation Issues with Window-based Protocol

- Receiver window size: # of out-of-sequence packets that the receiver can receive
- Sender window size: # of total outstanding packets that sender can send without acknowledged
- How to deal with sequence number wrap around?

What is Used in Practice?

- No flow or error control.
  - E.g. regular Ethernet, just uses CRC for error detection
- Flow control only.
  - E.g. Gigabit Ethernet
- Flow and error control.
  - E.g. X.25 (older connection-based service at 64 Kbs that guarantees reliable in order delivery of data)